

(t, r) Zero Forcing

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Standard Zero Forcing: Core Idea

Let $G = (V, E)$ be a simple graph.

- ▶ Each vertex is colored either black (active) or white (inactive).
- ▶ Start with an initial black set $B_0 \subseteq V$.
- ▶ Apply the color-change rule repeatedly.

Color change rule: If a black vertex u has

Process and Terminology

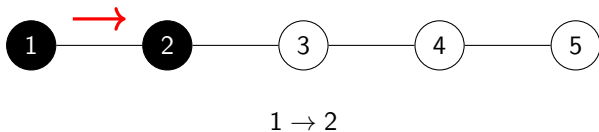
- ▶ A sequence of valid forces is a forcing process.
- ▶ If every vertex eventually becomes black, then B_0 is a zero forcing set.
- ▶ The minimum size of such a set is the zero forcing number:

Animated Example on the Path P_5

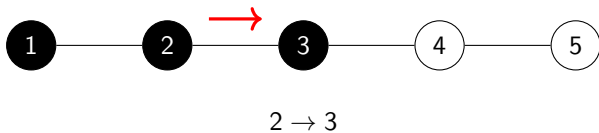


$$B_0 = \{1\}$$

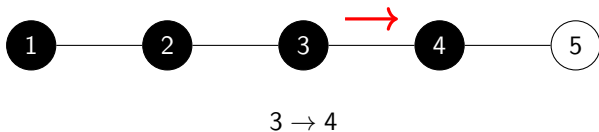
Animated Example on the Path P_5



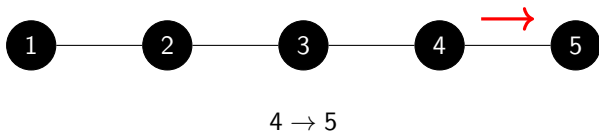
Animated Example on the Path P_5



Animated Example on the Path P_5



Animated Example on the Path P_5



Animated Example on the Path P_5

All vertices are black, so $\{1\}$ is a zero forcing set for P_5 .

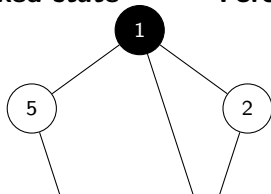
Why the Path Example Works

- ▶ Start with one black endpoint.
- ▶ At each step, the current frontier vertex has exactly one white neighbor.
- ▶ This keeps creating

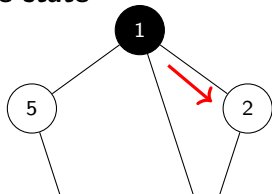


Cycle With a Chord: Not Every Move is Legal

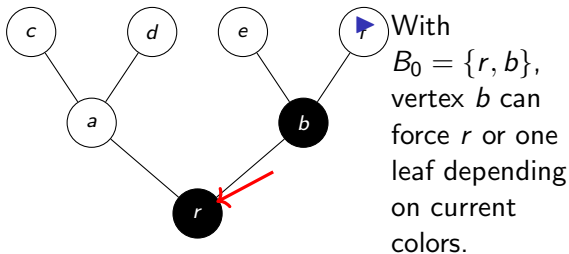
Blocked state



Forceable state



Tree Example: Initial Choice Matters



With $B_0 = \{r, b\}$, vertex b can force r or one leaf depending on current colors.

- If the wrong vertices are

Takeaway and Transition

- ▶ Standard zero forcing tracks deterministic propagation via the unique-white-neighbor rule.
- ▶ The central optimization problem is minimizing $|B_0|$, giving $Z(G)$.
- ▶ This classical process is our baseline for (t, r) zero forcing, where force dynamics